MMC TECHNICAL PROPOSAL KIT · ENGLISH EDITION

Thirteen Tender Documents, One Set of Numbers

Fillable, printable and internally reconciling. The take-off feeds the price schedule, the price schedule feeds the payment ladder, and the last review catches the inconsistency before the client does.

DOCUMENTS	GUIDES
13	2
SHEETS	**VERSION**
2	2.0

> **Read this page, then open TP-G2 first.** The kit is not thirteen independent forms — it is one offer expressed thirteen ways. Filling them out of sequence is how offers end up saying three different things about the same total.

01 Document Catalogue

CODE	DOCUMENT	ROLE	SCOPE
TP-01	**Cover Letter & Compliance Statement**	SUBMISSION	Signed undertaking plus a twelve-line compliance matrix mapped to the tender requirements.
TP-02	**Scope Understanding & Methodology**	TECHNICAL	Nine work packages, six-step execution sequence, coordination and clash management.
TP-03	**Time Schedule & Payment Schedule**	TECHNICAL	Fourteen-activity bar chart over twelve months, six milestones and the payment ladder.
TP-04	**QA/QC Plan**	TECHNICAL	Twelve-point inspection and test plan with hold points, NCR procedure and KPI targets.
TP-05	**HSE Plan**	TECHNICAL	Ten critical hazards, PPE schedule, six permit types and emergency response.
TP-06	**Resource Plan**	TECHNICAL	Organisation chart, key personnel schedule, fourteen plant items, ten material sources.
TP-07	**Track Record & Similar Projects**	QUALIFICATION	Ten project slots, detail sheets, and an experience matrix against the stated minimums.
TP-08	**Risk Register & Mitigation**	TECHNICAL	Fourteen risks on a 5×5 matrix with triggers, owners and costed contingencies.
TP-09	**BOQ Summary & Price Schedule**	COMMERCIAL	Twenty-four items, waste per row, overhead and profit as visible lines, four currencies.
TP-10–13	**Supporting Forms**	ADMINISTRATIVE	Tender checklist, RFI, deviations and value engineering, handover warranty.
TP-G1	**Drawing Reading Guide**	GUIDE	Architectural, structural and MEP: symbols, bar schedules, risers, cross-checks.

CODE	DOCUMENT	ROLE	SCOPE
TP-14	**Quantity Take-off & Summary**	COMMERCIAL	Dimensions in, quantities out: nine work groups, waste by material, live totals.
TP-G2	**How to Use the Kit**	GUIDE	The order of work, the quantity-to-price chain and a fifteen-point pre-submission review.

02 The Chain That Has to Hold

TP-14
Measure
L × W × H × No., waste per material

TP-14
Summarise
Nine group subtotals

TP-09
Price
Rate per item, plus overhead and profit

TP-03
Distribute
Payment % per milestone

TP-01
Declare
One figure, stated once

Two things you must change before submitting

The rates. Every unit rate in TP-09 is illustrative — it demonstrates the arithmetic and is not a market price. Replace all twenty-four with live supplier or subcontract quotations.
The people and projects. Every CV, client and reference in TP-06 and TP-07 must be real and verifiable. A reference the evaluator cannot reach is worse than no reference.

03 Using the Pages

- **Type directly into any field.** Entries save to your own browser as you type; nothing is transmitted anywhere.
- **Print or save as PDF** from the button in the top bar. A4, with background graphics enabled.
- **Switch currency** between USD, JOD, SAR and AED in the top bar. Conversion happens at display time, so nothing you typed is lost.
- **Edit a measurement or a rate** on TP-09 or TP-14 and every dependent total recalculates.

04 Also Available

OVERVIEW

What the kit is and who it is for

The problem it solves, the documents it contains, and what the pages look like.

PRICING

Starter, Professional, Enterprise

Monthly or annual, with the reasoning behind where the tiers are drawn.

COMPARE

Full feature matrix

Exactly what differs between the three plans, with nothing hidden in an add-on.

FAQ

Questions contractors ask first

The forms, the numbers, billing, data privacy, editions and support.

FORM TP-01 · COVER LETTER & COMPLIANCE STATEMENT

Cover Letter and Compliance & Undertaking Statement

Print on company letterhead and sign by the authorised representative. This is the only document the evaluation committee reads before it opens anything else.

TENDER NO.	PROJECT
—	—
SUBMISSION DATE	COPIES
—	**Original + 1**

> **Why this form decides the rest:** an unsigned cover letter, a missing bid bond reference or a qualification hidden in a footnote is an administrative rejection before technical scoring begins. Complete every field, and list every deviation in Form TP-12 rather than in this letter.

1 Bidder Particulars

REGISTERED COMPANY NAME *

As on trade licence

LEGAL FORM

LLC / PLC / Sole proprietorship

COMMERCIAL REGISTRATION NO.

TAX REGISTRATION / VAT NO.

CONTRACTOR CLASSIFICATION / GRADE

e.g. Grade 1 — Buildings

CONTRACTORS' ASSOCIATION MEMBERSHIP NO.

REGISTERED ADDRESS

TELEPHONE / EMAIL

AUTHORISED REPRESENTATIVE (NAME & TITLE) *

POWER OF ATTORNEY / DELEGATION REF.

2 Tender Particulars

EMPLOYER / CLIENT *

PROJECT NAME AS IN THE TENDER DOCUMENTS *

TENDER / RFQ NUMBER

CONDITIONS OF CONTRACT REFERENCE & DATE

SUBMISSION DEADLINE

REQUIRED CONSTRUCTION PERIOD

e.g. 12 months

OFFER VALIDITY PERIOD

e.g. 90 days

BID BOND VALUE & ISSUER

PERFORMANCE BOND OFFERED

% of contract value

3 Letter

> **Subject:** Submission of Technical and Financial Proposal for ..... — Tender No. .....

Dear Sir / Madam,

Having examined the Instructions to Bidders, the General and Particular Conditions of Contract, the Specifications, the Bill of Quantities, the Drawings and all addenda issued in respect of the above tender, and having visited the site and obtained all information necessary for the preparation of this offer, we, the undersigned, submit our technical and financial proposal for the execution and completion of the Works and the remedying of any defects therein.

We confirm that we have read and understood the complete tender documents, including every addendum and clarification issued, and that this offer is based on a full understanding of the scope of work and of the site conditions prevailing at the location of the Works.

We undertake to execute and complete the Works in accordance with the Specifications, Drawings and Bill of Quantities, to the quality standard required, for the total tender sum and within the programme stated in the financial offer and the time schedule appended hereto, and in any event within the period stated in the Instructions to Bidders.

We confirm that all documents and information submitted with this offer are true and accurate, and that any misstatement or concealment entitles the Employer to reject the offer or to take such other action as the contract permits.

This offer shall remain valid for the period stated in Section 2 above from the deadline for submission, and we undertake to furnish a performance security in the amount and form and on the terms set out in the Conditions of Contract should the award be made to us.

We understand that the Employer is not bound to accept the lowest or any offer, and that no cost incurred in the preparation of this offer will be reimbursed.

We remain at your disposal for any clarification or clarification meeting you consider appropriate.

Yours faithfully,

Name of authorised signatory

Name — block capitals

Title / position

Position

Date

Date

Company stamp

Company stamp

4 Compliance & Undertaking Statement

State the bidder's position against each heading. Any departure from the specification must be recorded in the Deviations schedule (Form TP-12) and cross-referenced here — a departure that appears only in this table and not in TP-12 will be treated as an unqualified offer.

#	REQUIREMENT IN THE TENDER DOCUMENTS	POSITION	REFERENCE / REMARKS
1	Full compliance with the general and particular technical specifications	COMPLY / COMPLY W/ NOTE / DEPART	
2	Compliance with the Bill of Quantities without altering items or units	COMPLY / COMPLY W/ NOTE / DEPART	
3	Compliance with the stated construction period and programme requirements	COMPLY / COMPLY W/ NOTE / DEPART	
4	Compliance with quality control, inspection and testing requirements	COMPLY / COMPLY W/ NOTE / DEPART	
5	Compliance with health, safety and environmental requirements	COMPLY / COMPLY W/ NOTE / DEPART	
6	Acceptance of the general and particular conditions of contract without qualification	COMPLY / COMPLY W/ NOTE / DEPART	

#	REQUIREMENT IN THE TENDER DOCUMENTS	POSITION	REFERENCE / REMARKS
7	Provision of the warranties, guarantees and insurances required	COMPLY COMPLY W/ NOTE DEPART	
8	Compliance with any stated material brands, rates or provisional sums	COMPLY COMPLY W/ NOTE DEPART	
9	Acceptance of the stated period for approval of materials and shop drawings	COMPLY COMPLY W/ NOTE DEPART	
10	Acceptance of the defects liability and warranty period	COMPLY COMPLY W/ NOTE DEPART	
11	Acceptance of the liquidated damages regime stated in the contract	COMPLY COMPLY W/ NOTE DEPART	
12	Confirmation that the tender sum includes all duties, taxes and levies applicable	COMPLY COMPLY W/ NOTE DEPART	

General declaration

We, the undersigned, declare that all information, data and documents contained in this offer are true and accurate; that we possess the technical, financial and organisational capability required to execute the Works; and that we have not knowingly included any misleading or incomplete information intended to affect the award decision. We understand that any breach of this declaration exposes us to legal liability and to exclusion from the tender.

NAME	POSITION	SIGNATURE	DATE

FORM TP-02 · SCOPE UNDERSTANDING & METHODOLOGY

Scope Understanding and Methodology of Execution

Demonstrates that the bidder has read the project, not only the Bill of Quantities.
Evaluators score this against the actual scope, so specific beats generic every time.

SECTIONS	WORKS PACKAGES
5	9

SEQUENCING	REV.
By zone & floor	A

> **Rule for this form:** every paragraph must name something that exists in this project — a zone, an element, a constraint. A methodology that could be pasted into any other tender scores zero, because the evaluator cannot verify that you understood *this* one.

1 Scope Understanding Summary

CONTRACT SCOPE (IN BRIEF)

EXCLUDED / BY OTHERS

Owner-supplied items, utilities diversions…

DISCIPLINE	INCLUDED (✓)	SCOPE NOTES / KEY QUANTITIES	PRINCIPAL RISK
Earthworks & site preparation			
Concrete substructure			
Concrete superstructure			
Structural steel / precast			
Masonry & blockwork			
Plaster, paint & finishes			
Tiling & flooring			
Joinery, doors & windows			
Roofing & waterproofing			
Electrical installations			
Mechanical & plumbing			
HVAC / ventilation			
Fire detection & protection			
Lifts / vertical transport			

DISCIPLINE	INCLUDED (✓)	SCOPE NOTES / KEY QUANTITIES	PRINCIPAL RISK
External works & landscaping			
Testing, commissioning & handover			

2 Overall Execution Methodology

We execute the Works on a **zone-by-zone, floor-by-floor** basis in which each zone passes through the same four gates before the next trade enters it: setting-out release, structural sign-off, services first-fix completion, and finishing release. This keeps trades from overtaking each other and gives the supervision team a single, repeatable checkpoint per zone rather than a project-wide free-for-all.

1. **Mobilisation and setting-out release**

   Site establishment, temporary facilities, survey control transfer from the Employer's benchmarks, and a joint setting-out check signed by the site engineer and the consultant before any excavation begins. No excavation starts on an unsigned setting-out sheet.

2. **Substructure sequence**

   Excavate and blind in panels, not the full footprint, so that reinforcement and formwork crews work continuously rather than in stop-start waves. Each panel is released for concreting only against a signed reinforcement inspection and a concrete pour card.

3. **Superstructure cycle**

   A fixed seven-day cycle per floor: day 1 column steel, day 2 column shutters and pour, day 3–4 beam and slab soffit, day 5 beam and slab steel, day 6 services sleeves and embedments, day 7 slab pour. The cycle is what makes the programme credible; it is measured weekly and reported as a variance.

4. **Envelope and weathertightness**

   Envelope closes zone by zone so that internal finishes follow behind a dry line rather than waiting for the whole building. Waterproofing is flood-tested before screed, and the test record is a hold point.

5. **Services first and second fix**

   Coordinated services drawings are produced from the tender documents before any first fix. Sleeves and box-outs go in with the structure; chasing of completed concrete is prohibited without written approval, because it is the single most common cause of latent structural complaints.

6. **Finishes and handover**

   Finishes follow a *mock-up first* rule: one room or one typical bay of every finish type is completed, inspected and approved as the benchmark before the trade is released to the wider area.

3 Methodology by Principal Work Package

3.1 Earthworks and foundations

Excavation to the depths shown, with sides battered or supported according to the soil report and the adjacent-structure survey. Dewatering is by perimeter sumps and pumps discharging to a silt trap before any connection to the storm network. Backfill is placed in 200 mm layers, each compacted and tested for field density at the frequency stated in the specification; a layer that has not passed is not covered.

3.2 Concrete works

Concrete is supplied from a ready-mix plant with an approved mix design per strength class. Every delivery is checked for slump, temperature and age before discharge, and cubes are taken at the specified frequency with a register that ties each cube set to a pour reference and a structural element. Formwork is designed for the pour pressure actually expected, not by rule of thumb, and struck only when the cube results or the maturity record permit it.

3.3 Reinforcement

Bar bending schedules are produced from the structural drawings and issued to the supplier; the site checks delivered bar against a cutting list, not against the drawing. Cover is controlled with approved spacers at the specified density — cover failures are a durability defect, not a cosmetic one, and they are almost always a spacer-density failure.

3.4 Masonry and blockwork

Blockwork is set out from the structural grid with movement joints at the locations shown. Lintels and bearing details follow the structural drawings. Wall chases are cut by machine to a fixed depth, never by hammer, and no chase runs horizontally across a wall for more than the permitted length.

3.5 Finishes

Plaster is applied to a mechanically prepared background with a bonding agent where the substrate requires it, and checked for flatness with a 2 m straightedge at the stated tolerance. Painting follows the full specified system — primer, intermediate and finish coats — with the dry film thickness verified at each coat; a two-coat job described as three is a defect that shows up within two seasons.

3.6 Electrical

Containment is installed before cable, cable before accessories, and accessories before energisation. Every circuit is identified at both ends and recorded in the cable schedule; insulation resistance and continuity tests are recorded per circuit, and the board schedule is issued as a document, not left as a handwritten card inside the panel.

3.7 Mechanical, plumbing and HVAC

Pipework is pressure-tested in sections before concealment, with the test pressure, duration and drop recorded on a signed test sheet. Drainage is air-tested and then flow-tested. Ductwork is leak-tested to the class stated, and air balancing is carried out by an independent testing and balancing agency whose report is the acceptance document for the HVAC system.

3.8 Fire protection

Sprinkler and standpipe pipework is hydrostatically tested and flushed before connection to the riser. Fire alarm loops are tested device by device against the cause-and-effect matrix, and the matrix itself is verified with the fire consultant before the demonstration to the civil defence authority.

3.9 External works

Levels are set from the finished floor level of the building, not from the road, so that the building drains away from itself. Paving falls, kerb lines and the tie-in to the public network are surveyed and signed off before the final surface course is placed.

4 Coordination and Clash Management

Services coordination is a scheduled activity with its own deliverables, not something that happens when a pipe hits a beam:

- A combined services model or, at minimum, coordinated overlays at 1:50 for plant rooms, risers, corridors and ceiling voids — the four places where clashes actually occur.
- Clash reports issued weekly, each with an owner, a decision and a closing date. An open clash older than fourteen days is escalated to the project manager in writing.
- Penetrations and sleeves agreed with the structural engineer before the pour, never after.
- Ceiling void depth reserved for services at the design stage and protected during installation; ceiling levels are set after the services are installed, not before.

5 Methodology Commitments

#	COMMITMENT	MEASURE OF COMPLIANCE
1	One methodology statement per work package, issued before the package starts	Approved method statement on file before first mobilisation
2	No pour without a signed reinforcement and formwork inspection	Pour card register, 100% coverage
3	No concealment without a signed services test	Test sheet per section, cross-referenced to location
4	Mock-up approval before any finish trade proceeds	Signed mock-up acceptance per finish type
5	Weekly programme variance reported, not discovered at the end	Variance report issued every week
6	Every deviation from this methodology approved in writing by the consultant	Approved technical query on file

Prepared by	Reviewed by	Approved by
Technical manager	Project manager	General manager

FORM TP-03 · TIME SCHEDULE, MILESTONES & PAYMENT SCHEDULE

Time Schedule, Milestones and Payment Schedule

A bar chart the evaluator can read in ten seconds, plus the payment ladder that has to match it. If the percentages here do not match the BOQ sections, the offer is internally inconsistent.

CONTRACT PERIOD	ZONES
12 months	**3**

MILESTONES	REV.
6	**A**

> **Consistency test:** the value of work complete at each milestone must equal the cumulative payment percentage claimed at that milestone, and both must be defensible from the BOQ. Three documents — programme, payment schedule and BOQ summary — are read together by every competent evaluator.

1 Schedule Particulars

CONTRACT PERIOD

12 months

START DATE / POSSESSION OF SITE

COMPLETION DATE *

WORKING DAYS PER WEEK

6

MOBILISATION PERIOD

30 days

DEFECTS LIABILITY PERIOD

12 months

WEATHER ALLOWANCE

days

CONTINGENCY IN PROGRAMME

%

2 Bar Chart (Gantt)

Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Mobilisation & site establishment Temporary facilities, survey, HSE set-up												
Setting-out & excavation Foundations, tie beams, dewatering												

Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Blinding & substructure concrete Footings, tie beams, backfill												
Superstructure — frame Columns, beams, slabs, 7-day cycle												
Blockwork & internal walls Follows structure two floors behind												
MEP first fix Containment, sleeves, pipework in slabs												
Envelope & waterproofing Roof, external walls, wet areas												
MEP second fix Cable, pipework, ductwork, terminals												
Plaster & screed Wet trades complete before finishes												
Tiling, painting & joinery Mock-up release per finish type												
External works & landscaping Paving, drainage tie-in, kerbs												
Testing & commissioning TAB, fire alarm, electrical, integrated test												
Snagging & handover Punch list, O&M, training, keys												
Defects liability period 12 months from substantial completion												

Bar chart is indicative of sequence and overlap; the contract programme issued after award is a critical-path network with resource loading.

3 Planned Monthly Progress

MONTH	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	TOTAL
Planned cumulative %	4	11	19	30	42	53	63	74	85	93	98	100	100
Planned value (USD)													

Fill the planned value row from the BOQ summary so that the cash curve, the programme and the payment schedule are three views of one plan.

4 Milestones and Payment Schedule

#	MILESTONE	COMPLETION CRITERION	CUM. %	PAYMENT %	RETENTION %
1	Mobilisation complete	Site facilities operational, HSE plan approved, survey transferred	4%	4%	10%
2	Substructure complete	All footings and tie beams cast, backfill tested and approved	19%	15%	10%
3	Superstructure complete	Roof slab cast, structure signed off by the structural engineer	53%	34%	10%

#	MILESTONE	COMPLETION CRITERION	CUM. %	PAYMENT %	RETENTION %
4	Envelope & first fix complete	Building weathertight, MEP first fix tested in sections	74%	21%	10%
5	Substantial completion	Finishes complete, commissioning reports issued, CO applied for	93%	19%	10%
6	Handover	As-builts, O&M, warranties, training and keys delivered	100%	7%	10%

Retention and release

Retention is deducted at the stated rate from each interim payment until the cap stated in the contract is reached. Half is released on the certificate of substantial completion; the balance on the certificate of completion of the defects liability period, against the maintenance certificate.

5 Schedule Control

- **Weekly:** actual versus planned progress by activity, with the physical percentage measured against the agreed weighting — not against the contractor's own estimate.
- **Monthly:** updated programme issued with the interim payment application, showing the critical path and any float consumed.
- **On any delay event:** notice within the period stated in the contract, with a contemporaneous record. Delay noticed months later is usually lost regardless of cause.
- **Recovery:** a written recovery plan is issued within seven days of the programme showing negative float, naming the additional resource and the cost of that resource.

Prepared by	Reviewed by	Approved by
Planning engineer	Project manager	Contracts manager

FORM TP-04 · QUALITY ASSURANCE & QUALITY CONTROL PLAN

Quality Assurance and Quality Control Plan (QA/QC)

Who owns quality, which inspections are hold points, what happens when work fails, and how records prove it afterwards.

SECTIONS	HOLD POINTS
7	6
ITP ITEMS	**REV.**
12	A

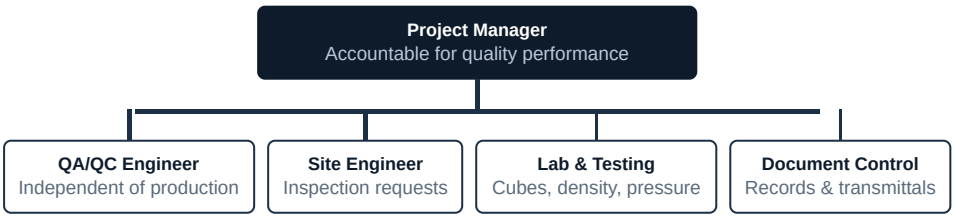
> **What evaluators look for:** not a policy statement, but a named quality engineer, a table of inspection and test points with the inspection authority marked, and a non-conformance procedure that has teeth. A plan without hold points is a hope, not a plan.

1 Quality Policy and Objectives

We deliver the Works to the specification, first time, and we prove it with records. The quality system on this project is a project-specific plan subordinate to our certified quality management system and is auditable against it. Conformance is verified by inspection and test before work is covered, not by retrospective argument after it is rejected.

#	OBJECTIVE	TARGET	VERIFICATION
1	First-time acceptance of structural inspections	≥ 95%	Inspection register
2	Non-conformances closed within 14 days	100%	NCR log
3	Concrete cube results meeting the specified class	≥ 97.5% at 28 days	Cube test register
4	Services pressure tests passed first attempt	≥ 90%	Test sheets
5	Snags per unit at handover	≤ 3	Snag list
6	Document transmittals acknowledged within 3 working days	100%	Transmittal log

2 Quality Organisation and Responsibilities



ROLE	QUALITY RESPONSIBILITY	AUTHORITY
Project Manager	Owns the QA/QC plan, chairs the weekly quality meeting, approves NCR disposition	Stops work on any non-conforming activity

ROLE	QUALITY RESPONSIBILITY	AUTHORITY
QA/QC Engineer	Issues inspection requests, witnesses hold points, maintains registers, raises NCRs	Rejects work; may not be overruled by production
Site Engineer	Raises inspection requests ahead of covering work, maintains as-built mark-ups	Requests inspection; cannot self-approve
Laboratory / third party	Samples, tests and reports to the specified standards	Reports independently to the consultant
Document Control	Registers, distributes and archives every record with a unique number	Refuses uncontrolled documents

3 Inspection and Test Plan (ITP)

H = hold point (work may not proceed without the consultant's witnessed approval) · **W** = witness point (notified; may proceed if the consultant does not attend) · **S** = surveillance (records reviewed).

#	ACTIVITY	INSPECTION / TEST	REFERENCE	TYPE	RECORD
1	Setting-out	Grid, benchmarks and levels verified against the survey	Drawings, survey data	H	Signed setting-out sheet
2	Excavation	Bottom level, bearing stratum, dimensions	Soil report, drawings	H	Inspection record
3	Blinding / formation	Level and cleanliness before reinforcement	Spec. §03300	W	Inspection record
4	Reinforcement	Bar size, spacing, laps, cover, chairs	BBS, structural drawings	H	Reinforcement checklist
5	Formwork	Line, level, plumb, tightness, release agent	Drawings	W	Formwork checklist
6	Concrete placement	Slump, temperature, age, cube sampling, vibration	Spec., mix design	H	Pour card + cube register
7	Backfill	Layer thickness, moisture, field density	Spec. §31200	W	Density test report
8	Blockwork	Alignment, plumb, mortar mix, movement joints, ties	Spec., drawings	S	Inspection record
9	Waterproofing	Surface prep, laps, flood test 24 h	Spec., manufacturer	H	Flood test certificate
10	MEP pressure tests	Test pressure, duration, drop	Spec., code	H	Signed test sheet
11	Electrical tests	Insulation resistance, continuity, earth loop	IEC 60364	W	Test certificate per circuit
12	Finishes mock-up	Flatness, adhesion, colour, workmanship	Spec., approved sample	H	Mock-up acceptance

4 Material and Procurement Control

- **Submittals before order:** material data sheets, certificates of conformity, samples and, where specified, independent test certificates are submitted for approval before any order is placed.

- **Approved brands only:** substitution requires written approval and a documented equivalence argument; an unapproved substitute installed is removed at our cost.
- **Delivery inspection:** every delivery is checked against the approved submittal and the delivery note, with damaged or non-conforming material segregated and tagged.
- **Storage:** cement and bagged materials off the ground and under cover; reinforcement on bearers; pipes capped; cable on drums off the ground. Storage damage is our risk, not a claim.
- **Traceability:** heat numbers on structural steel and batch numbers on concrete are recorded against the element they went into.

5 Non-Conformance (NCR) and Corrective Action

1. **Identify and segregate**

   Any non-conforming work is tagged and, where it affects safety or durability, stopped immediately by the QA/QC engineer. The tag stays until the disposition is closed.

2. **Raise the NCR**

   NCR number, date, location, description, specification clause breached, and the immediate containment action. Photographs are attached; the NCR is issued within 24 hours of identification.

3. **Disposition**

   One of: rework to specification, repair by an approved method, accept with the consultant's written concession, or reject and remove. Structural dispositions require the structural engineer's written agreement — the contractor cannot self-accept a structural deviation.

4. **Root cause**

   Stated in one sentence naming the process failure, not the person. "Spacer density not specified in the method statement" is a root cause; "operator error" is not.

5. **Corrective action**

   A change to the method statement, the checklist or the training record that prevents recurrence, with an owner and a date.

6. **Verify and close**

   Re-inspection confirms the fix; the closure is signed and the NCR log updated. An NCR closed without re-inspection is not closed.

NCR NO.	DATE	LOCATION / ELEMENT	DESCRIPTION	DISPOSITION	CLOSED	VERIFIED BY

6 Document and Record Control

Every record carries a unique number of the form `PROJECT-DISC-SEQ-REV` and is filed against the element it concerns, so that a future query about one column can be answered without searching the project. Records are retained for the period stated in the contract and handed over as part of the closeout document set: inspection records, test certificates, cube results, NCR log, material approvals and commissioning reports.

7 Internal Audit and Reporting

FREQUENCY	ACTIVITY	OUTPUT
Weekly	Quality walk with the consultant, open NCR review	Minutes and action list
Monthly	Internal audit of one process (rotating)	Audit report with findings
Monthly	Quality KPI report to the project manager	Objective dashboard
Quarterly	System review against the certified QMS	Management review record
At handover	Final record audit before the handover package is issued	Signed completeness certificate

Prepared by	Reviewed by	Approved by
QA/QC engineer	Project manager	Quality manager

FORM TP-05 · HEALTH, SAFETY & ENVIRONMENT PLAN

Health, Safety and Environment (HSE) Plan

Hazards identified for this project, the controls that actually remove them, and who is accountable when the control fails.

SECTIONS	CRITICAL HAZARDS
7	10
PERMITS	**REV.**
6	A

> **Scoring reality:** a generic HSE plan copied from another tender scores poorly because the hazards do not match the works. Name the actual high-risk activities on this project — deep excavation, work at height, lifting over live areas — and state the specific control for each.

1 HSE Policy and Responsibilities

No activity on this project is worth an injury. Every person on site has the authority to stop work they consider unsafe, without needing permission and without recrimination, and the stop-work record is reviewed by the project manager the same day rather than quietly buried.

ROLE	HSE RESPONSIBILITY	ACCOUNTABILITY
Project Manager	Provides resources, chairs the monthly HSE committee, authorises stop-work	Overall HSE performance
HSE Officer	Daily inspections, induction, permits, incident investigation, toolbox talks	Independent stop-work authority
Site Engineer	Implements controls at the work face, checks permits before work starts	Control effectiveness in his area
Foreman / supervisor	Enforces PPE, conducts pre-task briefings, reports hazards	Crew behaviour
Every worker	Complies, reports hazards and near misses, uses PPE	Own acts and omissions
Subcontractors	Comply with this plan; their own HSE plan must be equivalent, not inferior	Their own workforce and interfaces

2 Hazard Identification and Risk Assessment (HIRA)

Risk score = likelihood × severity on a 5×5 matrix. Residual risk is the score after the stated control is in place. Any residual score of 15 or above requires a written safe system of work approved by the project manager before the activity starts.

#	ACTIVITY	PRINCIPAL HAZARD	CONTROL MEASURES	RESIDUAL
1	Deep excavation	Collapse, engulfment, services strike	Battering or shoring to design; permit to excavate after service scan; edge protection at 1.1 m; ladder access every 7.5 m; no spoil within 1 m of edge	MEDIUM

#	ACTIVITY	PRINCIPAL HAZARD	CONTROL MEASURES	RESIDUAL
2	Work at height	Fall from edge or through opening	Guardrails and toe boards to all edges; openings covered and marked; scaffold tagged after inspection; harness and lanyard for MEWP work	MEDIUM
3	Lifting operations	Dropped load, crane overturn, crush	Approved lift plan for tandem or blind lifts; certified operator, banksman and rigger; exclusion zone under the load; no lift over occupied areas without a protected deck	MEDIUM
4	Concrete pours	Chemical burns, pump whip, noise	Gloves and goggles for all cement contact; pump line restrained and inspected; hearing protection within 85 dB; wash facilities on site	LOW
5	Electrical work	Electric shock, arc flash	Isolation and lock-out/tag-out; test before touch; no live work without a permit; RCD protection on all temporary supplies	MEDIUM
6	Confined spaces	Asphyxiation, toxic atmosphere	Entry permit; gas test before and during entry; continuous ventilation; attendant at the entry; rescue plan and equipment on standby	HIGH
7	Hot works	Fire	Hot work permit; 10 m clearance of combustibles; fire watch during and 60 min after; extinguisher at the work face	MEDIUM
8	Plant & vehicles	Struck by, run over	Segregated pedestrian routes; speed limit 10 km/h; reversing banksman; daily plant inspection	MEDIUM
9	Manual handling	Musculoskeletal injury	Mechanical aids for loads over 25 kg; team lifts; training in technique	LOW
10	Silica & dust	Respiratory disease	Wet cutting or extraction; FFP3 masks; no dry sweeping	MEDIUM

3 Personal Protective Equipment (PPE)

ITEM	WHEN MANDATORY	STANDARD
Safety helmet	At all times on site	EN 397
Safety footwear	At all times on site	EN ISO 20345 S3
High-visibility vest	At all times; Class 3 near plant	EN ISO 20471
Safety glasses	Cutting, grinding, drilling, chemical handling	EN 166
Gloves	Material handling, concrete, steel fixing	EN 388 / EN 374 for chemicals
Hearing protection	Any area above 85 dB(A)	EN 352
Respiratory protection	Dust, silica, solvent vapour	FFP3 / EN 140
Fall arrest harness	MEWP, leading edge, unprotected openings	EN 361 + EN 355 lanyard

4 Permit to Work System

No permit, no work. Permits are issued by the HSE officer for one shift only, are displayed at the work face, and are closed out by the issuer after inspection — not left to expire unattended.

PERMIT TYPE	REQUIRED FOR	ISSUED BY	VALIDITY
Excavation	Any excavation deeper than 1.2 m	HSE officer + site engineer	1 shift
Hot work	Welding, cutting, grinding, bitumen	HSE officer	1 shift
Confined space entry	Tanks, pits, manholes, ducts	Project manager + HSE officer	Per entry
Work at height	Above 2 m without permanent protection	HSE officer	1 shift
Lifting / critical lift	Tandem lifts, over live areas, over 80% capacity	Project manager + lift supervisor	Per lift
Electrical isolation	Any work on live or isolated systems	Authorised electrical person	1 shift

5 Training and Awareness

- **Site induction** before first entry, valid for the project, covering site rules, emergency arrangements, welfare and reporting. No induction, no access — enforced at the gate.
- **Task-specific training** for scaffolding, lifting, confined space, electrical work and plant operation, with certificates held on file and expiry dates tracked.
- **Toolbox talks** weekly, delivered by the supervisor at the work face, on the hazard of the week — not a repeat of the induction.
- **Emergency drills** every three months: evacuation, confined-space rescue and first aid response.

6 Emergency Response

ELEMENT	ARRANGEMENT
Assembly point	Marked, signed, and clear of plant routes; muster checked by roll call and headcount
Emergency contacts	Displayed at the gate, the site office and every floor: ambulance, civil defence, hospital, project manager
First aid	Two trained first aiders per shift; two kits, one at the office and one at the work face
Rescue	Confined-space and height rescue plan with equipment and a trained team on standby during the activity
Fire	Extinguishers at every hot work location and at plant; extinguisher training for all supervisors
Reporting	Any incident, however minor, reported within one hour; serious incidents reported to the authority within the statutory period

7 Incident Reporting and Investigation

INCIDENT DATE / TIME

LOCATION

CLASSIFICATION

Near miss / first aid / LTI / dangerous occurrence

DESCRIPTION AND IMMEDIATE ACTION TAKEN

What happened, who was involved, what was done at the time

ROOT CAUSE AND CORRECTIVE ACTION

Process failure, not a person; the change that prevents recurrence, with owner and date

METRIC	TARGET	REPORTED
Lost time injury frequency rate (per million hours)	< 1.0	Monthly
Recordable incidents	Zero	Monthly
Near misses reported	≥ 10 per month (a rising count is a healthy reporting culture)	Monthly
Toolbox talks delivered	100% of planned	Weekly
Open corrective actions older than 30 days	Zero	Weekly

Prepared by	Reviewed by	Approved by
HSE officer	Project manager	Corporate HSE manager

FORM TP-06 · RESOURCE PLAN

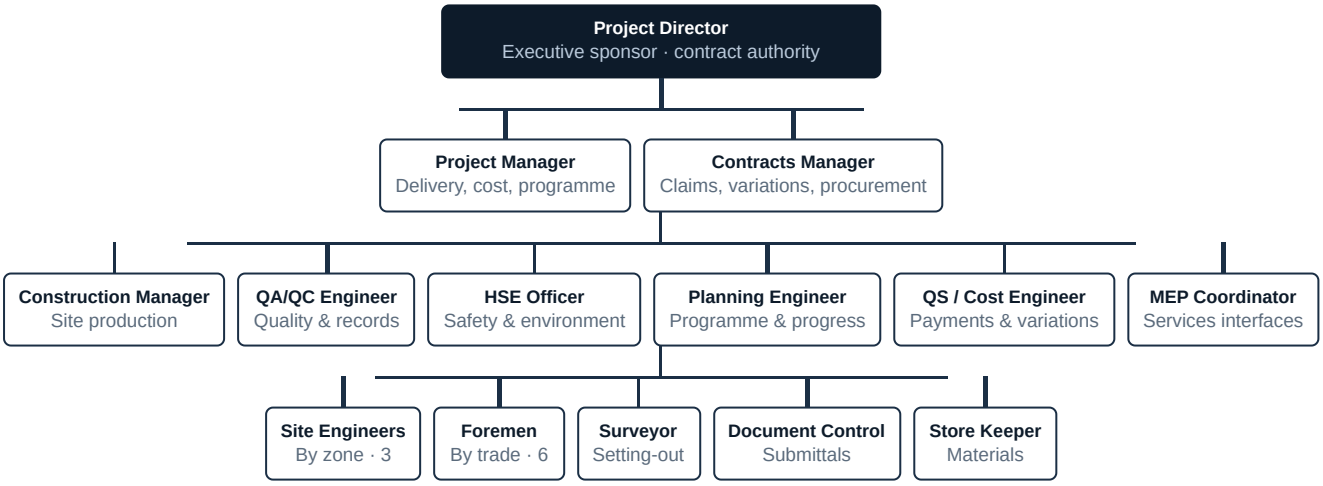
Resource Plan: Organisation, Key Personnel, Equipment and Materials

The evaluator is asking one question: can this bidder actually staff, plant and supply this project? Named people, owned plant and identified sources answer it; general assurances do not.

SECTIONS	KEY STAFF
5	8
PLANT ITEMS	**REV.**
14	A

> **What earns marks:** staff named with qualifications and years of experience, plant listed as owned or leased with capacity, and material sources named with an approved-supplier route. A CV that says “civil engineer” without a registration number is a CV nobody can score.

1 Project Organisation Chart



2 Key Personnel Schedule

#	POSITION	NAME	QUALIFICATION	YEARS EXP.	SIMILAR PROJECTS	ALLOCATION
1	Project Director					20%
2	Project Manager					100%
3	Construction Manager					100%
4	QA/QC Engineer					100%
5	HSE Officer					100%
6	Planning Engineer					60%

#	POSITION	NAME	QUALIFICATION	YEARS EXP.	SIMILAR PROJECTS	ALLOCATION
7	QS / Cost Engineer					50%
8	MEP Coordinator					80%
9	Site Engineer (x3)					100%
10	Surveyor					50%

Allocation is a commitment

A person shown at 100% cannot be shown at 100% on another concurrent project in the same offer. " Evaluators do check this, and a key person double-booked across two tenders is a credibility failure that " taints the whole submission.

3 Abbreviated CVs of Key Personnel

Curriculum Vitae — Key Person 1

NAME

POSITION ON THIS PROJECT

PROFESSIONAL REGISTRATION NO.

YEARS OF EXPERIENCE

ACADEMIC QUALIFICATION

CURRENT EMPLOYER / RELATIONSHIP

RELEVANT PROJECT EXPERIENCE (LAST 10 YEARS)

PROJECT	ROLE	VALUE	YEAR	SCOPE

LANGUAGES · AVAILABILITY · DECLARATION

Availability from (date); declaration that the CV is accurate and the person is available for this project

Repeat this block for each key person, or attach CVs in the same format as an annex and reference them here.

4 Equipment Schedule

#	EQUIPMENT	CAPACITY / SPEC	QTY	OWNED / LEASED	CONDITION	ON SITE FROM
1	Hydraulic excavator 20 t	0.9 m³ bucket	2	Owned	< 5 yrs	M1
2	Backhoe loader	JCB 3CX class	1	Owned	< 5 yrs	M1
3	Mobile crane 50 t	Telescopic boom	1	Leased	Certified	M4
4	Tower crane	Max 2.5 t @ 50 m	1	Leased	Certified	M4
5	Concrete pump	Truck-mounted 36 m	1	Leased	Certified	M3
6	Transit mixer 8 m³	With agitator	4	Leased	< 5 yrs	M3
7	Vibrator	Poker 50 mm	8	Owned	Good	M3
8	Plate compactor	90 kg	4	Owned	Good	M2
9	Generator 250 kVA	Silenced	2	Owned	Good	M1
10	Water pump 75 mm	Diesel	6	Owned	Good	M2
11	Scaffolding	Cuplock, 3,000 m²	1 set	Owned	Certified	M5
12	MEWP 16 m	Articulated boom	1	Leased	Certified	M6
13	Survey total station	2 second accuracy	2	Owned	Calibrated	M1
14	Laser level	Self-levelling	4	Owned	Calibrated	M1

5 Material Sources

MATERIAL	SOURCE / SUPPLIER	LOCATION	APPROVAL ROUTE	LEAD TIME
Ready-mix concrete		Batching plant	Mix design + trial mix	3 days
Reinforcement steel		Mill / stockist	Mill certificates + sample	14 days
Cement		Distributor	Certificate of conformity	7 days
Blockwork		Local factory	Sample + test certificate	10 days
Structural steel		Fabricator	Shop drawings + mill certs	30 days
Waterproofing		Approved applicator	Manufacturer warranty	10 days
Tiles & sanitaryware		Importer	Sample board	30 days
Electrical cable & boards		Distributor	Type test certificates	21 days
HVAC equipment		Manufacturer	Performance data + sample	45 days
Fire protection		Approved supplier	Civil defence listing	30 days

Prepared by
Planning engineer

Reviewed by
Project manager

Approved by
General manager

FORM TP-07 · TRACK RECORD & SIMILAR PROJECTS

Track Record and Similar Projects

Similar means similar in type, size, complexity and delivery model — not merely large.
Each entry must be verifiable by a third party without contacting the bidder.

SECTIONS	PROJECT SLOTS
5	10
VERIFICATION	**REV.**
Certificates	A

> **Test for every entry:** would the evaluator accept this as relevant to the works being tendered? If the project is a different building type, state the transferable element explicitly — the same structural system, the same MEP complexity, the same live-environment constraints.

1 Company Profile

YEAR ESTABLISHED

CONTRACTOR CLASSIFICATION / GRADE

NUMBER OF PERMANENT STAFF

ANNUAL TURNOVER — LAST YEAR

ANNUAL TURNOVER — YEAR BEFORE

LARGEST SINGLE CONTRACT COMPLETED

VALUE OF WORK CURRENTLY IN HAND

BONDING CAPACITY AVAILABLE

GEOGRAPHIC AREAS OF OPERATION

2 Similar Projects Completed

#	PROJECT & LOCATION	CLIENT / CONSULTANT	TYPE	CONTRACT VALUE	YEAR	DURATION	ROLE
1							Main contractor
2							Main contractor
3							Main contractor
4							Main contractor
5							Main contractor
6							Joint venture
7							Main contractor
8							Main contractor
9							Main contractor
10							Main contractor

3 Project Detail Sheets

Complete one sheet for each of the three most relevant projects. " These are the entries the evaluator will actually read.

PROJECT NAME

LOCATION

CONTRACT VALUE

CLIENT

CONSULTANT / ENGINEER

CONTRACT FORM

FIDIC Red Book / national form

START / COMPLETION

ACTUAL VS CONTRACT DURATION

FINAL ACCOUNT VS CONTRACT SUM

BUILT-UP AREA

m²

FLOORS / HEIGHT

STRUCTURAL SYSTEM

SCOPE OF WORKS EXECUTED BY THE BIDDER

Disciplines, packages and any specialist subcontracts managed

WHY THIS PROJECT IS RELEVANT TO THE PRESENT TENDER

Name the specific similarity: structural system, MEP complexity, live environment, phasing, size band

CLIENT REFERENCE (NAME, POSITION, TELEPHONE, EMAIL)

A reference the evaluator can actually reach

4 Experience Compliance

REQUIREMENT IN THE TENDER	STATED MINIMUM	BIDDER'S POSITION	EVIDENCE REFERENCE
Years of experience in similar works			
Number of similar projects completed			
Largest single contract value			
Annual turnover threshold			
Key personnel experience			
Local registration / classification			
ISO 9001 certification			
ISO 45001 / OHSAS certification			
ISO 14001 certification			

5 Certificates and References

- Certificates of completion, taking-over or substantial completion for each project cited — not merely a contract award letter.
- Client satisfaction or performance letters where available; a reference letter that names the project, the value and the completion date is worth more than one that does not.
- Valid ISO certificates with accreditation body and expiry date, plus the scope statement — a certificate whose scope does not cover building construction proves nothing.
- Contractor classification and trade licence, current and legible.

DOCUMENT	ISSUER	REFERENCE NO.	DATE / EXPIRY	ATTACHED

Prepared by

Contracts manager

Verified by

Project director

FORM TP-08 · RISK REGISTER & MITIGATION PLAN

Risk Analysis and Mitigation Plan

Risks the bidder has actually priced and planned for. A register that lists only force majeure tells the evaluator the bidder has not thought about this project.

SECTIONS	RISKS
4	14

MATRIX	REV.
5×5	A

Scoring logic: each risk needs an owner, a trigger that tells you it is happening, a mitigation that acts before the impact, and a contingency with a value and a time allowance. “Monitor closely” is not a mitigation.

1 Risk Assessment Methodology

Risk score = **likelihood × consequence**, each scored 1–5. Scores are reviewed at the weekly progress meeting and any movement of two points or more is reported to the project director.

	1 Insign.	2 Minor	3 Moderate	4 Major	5 Severe
5 Almost certain	5	10	15	20	25
4 Likely	4	8	12	16	20
3 Possible	3	6	9	12	15
2 Unlikely	2	4	6	8	10
1 Rare	1	2	3	4	5

BAND	SCORE	REQUIRED RESPONSE
CRITICAL	15–25	Project director decision; written plan before activity starts
HIGH	10–14	Project manager plan; weekly review
MEDIUM	5–9	Construction manager plan; monthly review
LOW	1–4	Accepted; monitored through routine reporting

2 Risk Register

#	RISK EVENT	CATEGORY	L	C	SCORE	MITIGATION (BEFORE IMPACT)	TRIGGER TO ACT	OWNER
1	Late site handover / possession delay	Client	3	4	12	Mobilisation plan with a 30-day compressed start; advance procurement of long-lead items against a letter of intent	Handover date missed by 7 days	PM
2	Ground conditions differ from the soil report	Technical	3	4	12	Pre-bid site visit and local knowledge; provisional allowance for rock excavation; early trial pit	Refusal rates in the first excavation panel	CM
3	Utility diversion delays excavation	External	3	4	12	Service scan and trial trenches before excavation permit; diversion application in month 1	Uncharted service found	CM
4	Design coordination gaps between disciplines	Design	4	3	12	Coordinated services model before first fix; weekly clash report with owners and dates	>5 open clashes older than 14 days	MEP Coord.
5	Late approval of materials or shop drawings	Client	4	3	12	Submittal register issued in week 1 with dates back-planned from the programme; escalate at day 10	Any submittal unreturned at 10 days	QA/QC
6	Long-lead equipment delivery slip	Supply	3	4	12	Order at award; factory acceptance test scheduled; alternative approved source identified	Delivery forecast slips 2 weeks	Procurement
7	Cash flow gap on interim payments	Financial	3	4	12	Payment application submitted on the contract date with full substantiation; retention and bond costs in the cash curve	Certification delayed beyond 21 days	Contracts
8	Price escalation on steel and cement	Market	3	3	9	Firm quotations for the first 6 months; escalation clause position stated in the offer	Index movement >5%	QS
9	Adverse weather stopping external works	External	3	3	9	Weather days in the programme; sequence external works in the dry window	3 consecutive lost days	Planner
10	Labour shortage for a specialist trade	Resource	3	3	9	Framework subcontractors with capacity commitments; second source identified per package	<80% of planned labour on site	CM

#	RISK EVENT	CATEGORY	L	C	SCORE	MITIGATION (BEFORE IMPACT)	TRIGGER TO ACT	OWNER
11	Safety incident causing a stop-work order	HSE	2	5	10	HSE plan with hold-point permits; daily inspection; stop-work authority for every worker	Any lost-time incident	HSE
12	Non-conformance requiring demolition and rebuild	Quality	2	4	8	ITP hold points; no pour without signed inspection; mock-up approval before finishes	Any NCR rated major	QA/QC
13	Scope creep through undocumented instructions	Contract	4	2	8	Written instruction required before work; daily site diary; variation register	Any verbal instruction received	Contracts
14	Handover delay from incomplete O&M and as-builts	Closeout	3	3	9	Closeout register opened at 50% completion, not at the end; documents tracked as line items	Any closeout item open at 85% progress	QA/QC

3 Critical Risks and Contingency

CRITICAL RISK	CONTINGENCY (TIME)	CONTINGENCY (COST)	TRIGGER FOR RELEASE
Ground conditions	10 days	Allowance in BO	Confirmation of rock or unsuitable material
Utility diversion	21 days		Service not diverted by the agreed date
Long-lead equipment	30 days		FAT date missed
Payment delay	—	Working capital	Certificate overdue by 30 days

4 Risk Management During Execution

- **Weekly:** register reviewed at the progress meeting; new risks added, closed risks archived with the reason for closure.
- **Monthly:** risk report to the Employer identifying movement, mitigation status and any risk that has become an issue requiring a decision.
- **Early warning:** any event likely to affect time or cost is notified in writing immediately, with a contemporaneous record — not reconstructed at the end of the project.
- **Escalation:** a critical risk is escalated to the project director within 24 hours, with a recommendation and the cost of each option.

Prepared by
Project manager

Reviewed by
Risk / contracts manager

FORM TP-09 · BOQ SUMMARY & PRICE SCHEDULE

Bill of Quantities Summary and Price Schedule

Quantities carried from the take-off sheet, priced per item, with waste, overhead, profit and tax shown as separate lines. Change the currency in the top bar and every figure converts; change a rate and every total follows.

ITEMS	GROUPS
24	9

OVERHEAD	PROFIT
12%	8%

How to price this form: enter the unit rate only. The amount, the group subtotals and the offer value are calculated. The rates currently shown are *illustrative* — they exist to demonstrate the arithmetic and must be replaced with live supplier and subcontract quotations before submission. The quantities are not illustrative: they are derived from the declared geometry of the worked example and can be verified by hand from the dimensions in each row.

1 Pricing Factors

OVERHEAD & SITE ESTABLISHMENT (%)

12

PROFIT (%)

8

TAX / VAT (%)

0

CURRENCY (DISPLAY)

Switch in the top bar

CONTINGENCY (%)

ESCALATION ALLOWANCE (%)

BOND & INSURANCE (%)

OFFER VALIDITY

90 days

2 Detailed Price Schedule

Net quantity = length × width × height × number. Gross quantity applies the waste percentage. Amount = gross quantity × unit rate.

CODE	DESCRIPTION	UNIT	L	W	H	NO.	NET	WASTE %	GROSS	AMOUNT	RATE
01 · EARTHWORKS & BACKFILL											
01-01	Excavation for foundations	m³	2.2	2.2	1.5	16	116.160	0	116.160	$ 1,068.67	9.20
01-02	Backfilling & compaction	m³	81.	1	1	1	81.296	10	89.426	$ 380.06	4.25
02 · CONCRETE WORKS											
02-01	Blinding concrete C15	m³	1.7	1.7	0.1	16	4.624	5	4.855	$ 582.62	120.00
02-02a	Isolated footings C30	m³	1.5	1.5	0.4	16	14.400	3	14.832	$ 2,343.46	158.00
02-02b	Tie beams C30	m³	0.3	0.6	88	1	15.840	3	16.315	$ 2,577.80	158.00
02-02c	Columns C30	m³	0.3	0.4	3.2	32	12.288	3	12.657	$ 1,999.75	158.00
02-02d	Beams C30	m³	0.2	0.5	88	2	22.000	3	22.660	$ 3,580.28	158.00
02-02e	Slabs C30	m³	12.	10.(	0.1	3	54.000	3	55.620	$ 8,787.96	158.00
03 · REINFORCEMENT											
03-01	High-yield reinforcement B420	t	11.	1	1	1	11.248	4	11.698	$ 11,405.72	975.00
04 · FORMWORK											
04-01	Formwork — sides, soffits and slabs	m²	893	1	1	1	893.760	5	938.448	$ 5,959.14	6.35
05 · MASONRY											
05-01	Lightweight blockwork	m²	522	1	1	1	522.480	6	553.829	$ 11,353.49	20.50
06 · PLASTER & PAINT											
06-01	Cement plaster	m²	166	1	1	1	1,668.960	8	1,802.477	$ 13,969.20	7.75
06-02	Emulsion paint	m²	166	1	1	1	1,668.960	5	1,752.408	$ 7,885.84	4.50
07 · TILING											
07-01	Floor tiles 600×600	m²	12.	10.(	1	2	240.000	10	264.000	$ 6,705.60	25.40
07-02	Wall tiles	m²	15.	1	1	12	180.000	10	198.000	$ 4,474.80	22.60
Total direct cost										**$ 117,014.31**	

CODE	DESCRIPTION	UNIT	L	W	H	NO.	NET	WASTE %	GROSS	AMOUNT	RATE
07-03	Skirting	m	15(	1	1	1	150.000	7	160.500	$ 2,038.35	12.70
08 · ELECTRICAL											
08-01	Lighting & socket points	no.	14	1	1	12	168.000	2	171.360	$ 5,312.16	31.00
08-02	Cable installations	m	18{	1	1	1	1,850.000	8	1,998.000	$ 10,689.30	5.35
08-03	Conduits & trunking	m	16(	1	1	1	1,600.000	8	1,728.000	$ 3,888.00	2.25
08-04	Distribution boards	no.	4	1	1	1	4.000	0	4.000	$ 820.00	205.00
09 · MECHANICAL & PLUMBING											
09-01	Copper A/C refrigerant pipework	m	22(	1	1	1	220.000	6	233.200	$ 4,104.32	17.60
09-02	PVC drainage	m	12(	1	1	1	120.000	7	128.400	$ 1,521.54	11.85
09-03	PPR water supply	m	26(	1	1	1	260.000	7	278.200	$ 2,434.25	8.75
09-04	Sanitary sets	set	12	1	1	1	12.000	0	12.000	$ 3,132.00	261.00
Total direct cost										**$ 117,014.31**	

3 Summary and Offer Value

WORK GROUP	AMOUNT	SHARE
Earthworks & backfill	$ 1,448.73	1.2%
Concrete works	$ 19,871.87	17.0%
Reinforcement	$ 11,405.72	9.7%
Formwork	$ 5,959.14	5.1%
Masonry & blockwork	$ 11,353.49	9.7%
Plaster & paint	$ 21,855.04	18.7%
Tiling & flooring	$ 13,218.75	11.3%
Electrical	$ 20,709.46	17.7%
Mechanical & plumbing	$ 11,192.11	9.6%
Total direct cost	**$ 117,014.31**	**100%**

BUILD-UP OF OFFER VALUE	RATE	AMOUNT
Direct cost (including waste)	—	$ 117,014.31
Overhead & site establishment	12.0%	$ 14,041.72
Subtotal after overhead	—	$ 131,056.03
Profit	8.0%	$ 10,484.48
Subtotal before tax	—	$ 141,540.51
Tax / VAT	0.0%	$ 0.00
Offer value	**—**	**$ 141,540.51**

Consistency checks before submission

- Group subtotals here must equal the Summary sheet totals in Form TP-14.
- The offer value must match the figure written in the financial offer and the cover letter.
- Payment percentages in TP-03 must be defensible from these group shares.
- Any group above 20% of the total deserves a second look at its rates.

4 Item Allocation by Stage

Links each work group to the milestone at which it is earned. " This is the bridge between the price schedule and the payment schedule.

WORK GROUP	MILESTONE	EARNED AT
Earthworks & backfill	M2	Substructure complete
Concrete works	M2–M4	Substructure and superstructure
Reinforcement	M2–M4	Substructure and superstructure
Formwork	M2–M4	Substructure and superstructure
Masonry & blockwork	M3–M6	Superstructure complete
Plaster & paint	M4–M5	Envelope & first fix complete
Tiling & flooring	M5	Substantial completion
Electrical	M4–M5	Substantial completion
Mechanical & plumbing	M4–M5	Substantial completion

Prepared by	Reviewed by	Approved by
Quantity surveyor	Commercial manager	General manager

FORMS TP-10 TO TP-13 · SUPPORTING FORMS

Supporting Forms: Tender Checklist, RFI, Deviations & Value Engineering, Handover Warranty

The four forms that prevent the most common causes of rejection, dispute and unpaid closeout claims.

FORMS	CHECKLIST ITEMS
4	22
SECTIONS	**REV.**
4	A

Sequence matters: fill TP-10 before you price, TP-11 and TP-12 before you submit, and TP-13 at handover. A deviations schedule raised after award is a variation; raised before submission it is a qualified offer you can defend.

TP-10 Tender Document Checklist

Tick each item before packing. A missing mandatory item is an administrative rejection that no technical score can recover.

#	DOCUMENT / REQUIREMENT	TYPE	STATUS	OWNER	NOTES
1	Signed and stamped cover letter	MANDATORY			
2	Bid bond / tender security	MANDATORY			
3	Commercial registration	MANDATORY			
4	Contractor classification certificate	MANDATORY			
5	Contractors' association membership	MANDATORY			
6	Social security clearance	MANDATORY			
7	Tax compliance certificate	MANDATORY			
8	Methodology and scope understanding (TP-02)	MANDATORY			
9	Time schedule and bar chart (TP-03)	MANDATORY			
10	QA/QC plan (TP-04)	MANDATORY			
11	HSE plan (TP-05)	MANDATORY			
12	Organisation chart and CVs (TP-06)	SCORED			
13	Equipment schedule (TP-06)	SCORED			
14	Track record and certificates (TP-07)	SCORED			
15	Risk register (TP-08)	SCORED			
16	Priced BOQ, all items and totals (TP-09)	MANDATORY			

#	DOCUMENT / REQUIREMENT	TYPE	STATUS	OWNER	NOTES
17	Financial offer letter	MANDATORY			
18	Deviations / qualifications schedule (TP-12)	SCORED			
19	Audited financial statements, 3 years	MANDATORY			
20	Bank reference / line of credit	SCORED			
21	ISO certificates with scope	SCORED			
22	Addenda acknowledgement	MANDATORY			

Before you seal the envelope

Count the copies, confirm the original is marked as such, check every signature and stamp, verify that the pricing in words and figures agree, and confirm the submission address and time. Late is late — there is no discretion, and a courier timestamp is not the Employer's clock.

TP-11 Request for Information / Clarification

RFI NUMBER

RFI-001

DATE RAISED

RESPONSE REQUIRED BY

DRAWING / SPECIFICATION REFERENCE

REVISION

DISCIPLINE

Arch / Struct / MEP / Civil

QUERY — STATE THE CONFLICT PRECISELY, WITH BOTH REFERENCES

e.g. Drawing S-201 rev C shows a 300x600 tie beam at grid B/2, while the BOQ item 02-02b describes 300x500. Confirm which governs for pricing and construction.

BIDDER'S PROPOSED RESOLUTION (OPTIONAL, BUT IT SHORTENS THE RESPONSE)

State what you propose and the cost/time effect if the Employer agrees

COST / TIME IMPACT IF UNRESOLVED

Quantify; an RFI with no impact statement is answered last

RFI NO.	SUBJECT	DATE RAISED	RESPONSE DUE	RESPONSE DATE	STATUS

RFI discipline

One question per RFI. A document with six questions gets one answer to three of them. Closing date for questions is stated in the instructions — an RFI raised after that date will not be answered and its subject becomes your risk at the price you bid.

TP-12 Deviations & Value Engineering

Record every departure from the specification, and every alternative you propose with its saving. An unlisted departure is deemed included at no extra cost.

A · DEVIATIONS

#	SPECIFICATION CLAUSE	REQUIREMENT	BIDDER'S PROPOSAL	REASON	COST EFFECT	TIME EFFECT
1						
2						
3						
4						

B · VALUE ENGINEERING PROPOSALS

#	ELEMENT	SPECIFIED SOLUTION	PROPOSED ALTERNATIVE	SAVING	EFFECT ON PERFORMANCE / WARRANTY
1					
2					
3					

How to make a VE proposal acceptable

A value engineering proposal is accepted on three conditions: it does not reduce performance or design life, it carries an equivalent or better warranty, and the saving is shared or passed on as the contract requires. A proposal that offers a saving but weakens the specification will be rejected and will damage the credibility of your other submissions.

TP-13 Quality & Handover Warranty

PROJECT NAME

CONTRACT NO.

SUBSTANTIAL COMPLETION DATE

DEFECTS LIABILITY PERIOD

12 months

WARRANTY START DATE

WARRANTY END DATE

ELEMENT / SYSTEM	WARRANTY PERIOD	PROVIDER	DOCUMENT REFERENCE	ATTACHED
Structural works	10 years (or as specified)	Contractor		
Waterproofing / roofing	10 years	Manufacturer + applicator		
External envelope / glazing	5 years	Manufacturer		
Mechanical / HVAC equipment	24 months	Manufacturer		
Electrical installations	24 months	Contractor		
Fire detection & protection	24 months	Manufacturer + contractor		
Lifts	24 months	Manufacturer		
Finishes	12 months	Contractor		
Joinery & ironmongery	12 months	Manufacturer		

UNDERTAKING

We undertake that the Works have been executed in accordance with the specification and the drawings, that all materials are new and of the specified quality unless otherwise agreed in writing, and that we will rectify at our own cost any defect arising from workmanship or materials during the defects liability period, within the response times stated in the contract. On expiry of that period, the manufacturer warranties listed above remain in force with the Employer as beneficiary.

DOCUMENT HANDED OVER	COPIES	FORMAT	DATE
As-built drawings	2 + PDF	DWG + PDF	
O&M manuals	2 + PDF	Hard + PDF	
Warranty certificates	1 set	Hard + PDF	
Test and commissioning reports	1 set	PDF	
Spare parts & attic stock	As scheduled	Physical	
Keys and access cards	As scheduled	Physical + register	
Training records	1 set	Signed attendance	

Contractor

Project manager

Employer / consultant

Authorised representative

GUIDE TP-G1 · DRAWING READING GUIDE

How to Read Architectural, Structural and MEP Drawings

A working sequence for reading a drawing set, the symbols that matter, and the details where quantity errors are actually made.

SECTIONS	DISCIPLINES
8	3
WORKED EXAMPLE	REV.
1	A

> **The rule that prevents most take-off errors:** read the sheet, then the schedule, then the detail, then the specification, then measure. Each of those five sources can contradict the one before it, and the contradiction is where the money is lost. A dimension scaled from a PDF is not a dimension.

1 Before You Read a Single Line

1. **Read the drawing register and the revision history**

   The register tells you what should exist; the revision block tells you what changed. A take-off from a superseded revision is the most expensive mistake in estimating, and it is silent — the sheet looks exactly like the current one.

2. **Confirm the revision you are measuring is the current issued revision**

   Check the revision letter on the sheet against the transmittal register, not against the file name. Files are renamed; drawings are not.

3. **Establish the scale and the units, and then do not scale**

   Most sets are in millimetres; some structural details are in centimetres. Measure only from the written dimensions; use the scale bar to check that a printed copy is not distorted.

4. **Identify the grid and the datum before anything else**

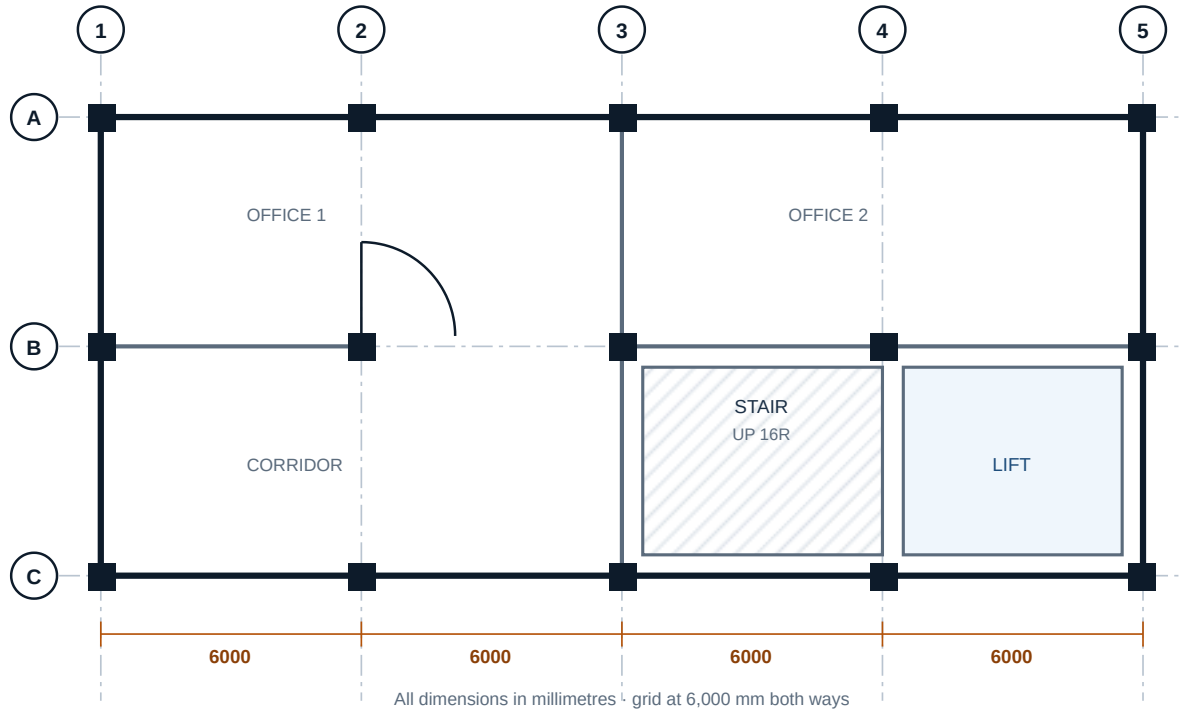
   Grid lines give you coordinates for every element; the datum gives you a single reference for every level. Without both, a slab thickness becomes a guess.

5. **Read the general notes**

   Concrete grades, cover, blockwork strengths, plaster thickness and paint systems are frequently stated only here — and they govern the whole take-off.

2 Architectural Drawings

Architectural drawings answer *what the building is*: spaces, dimensions, levels, finishes and openings. They are the source for blockwork, plaster, tiling, doors and windows — but only after the finishes schedule has been cross-checked, because room finishes are rarely written on the plan.



Typical floor plan — grid bubbles, column positions, wall thickness, door swing and dimension string. Read the grid first, then the wall centre-lines, then the openings.

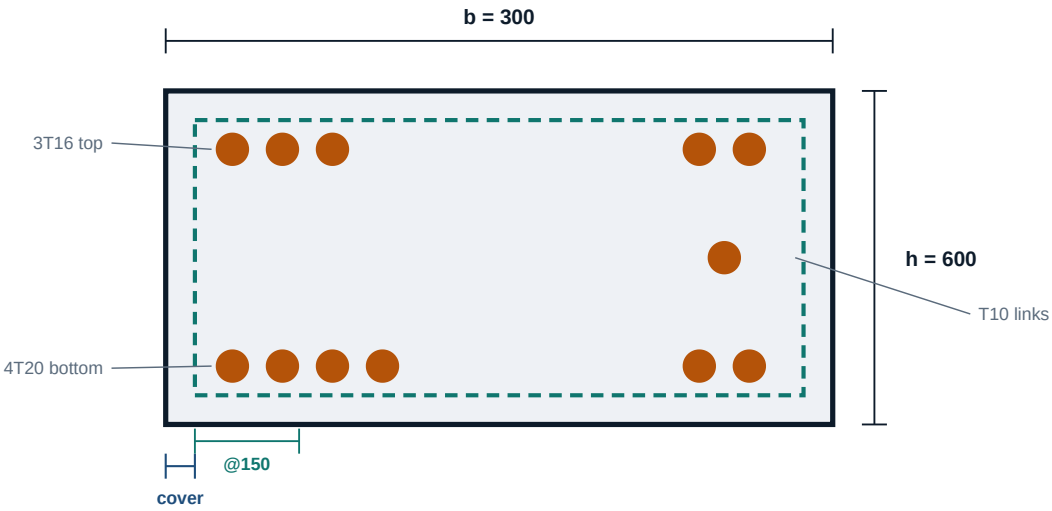
DRAWING TYPE	WHAT IT GIVES YOU	WHAT IT HIDES
Floor plan	Layout, room sizes, wall positions, door and window openings	Wall heights, finishes, structural thickness
Reflected ceiling plan (RCP)	Ceiling grid, levels, light fittings, diffusers, access panels	Ceiling void depth, which is where services clashes live
Elevations	Storey heights, façade composition, sill and head levels	Anything behind the façade
Sections	Floor-to-floor heights, slab and screed build-up, roof construction	Non-typical areas
Enlarged details	Thresholds, parapets, waterproofing laps, junctions	Continuity between details
Door and window schedule	Type, size, material, glazing, ironmongery, fire rating	Wall opening size, which may differ from the frame size
Finishes schedule	Floor, wall and ceiling finish per room	Areas — these must be measured from the plan
Room data sheets (if issued)	Every finish and fitting per room, plus MEP requirements	Nothing, when they exist — they are the best single source

Where architectural take-off goes wrong

Openings deducted twice, or not at all. Establish one rule — deduct openings above a stated area — and apply it to every wall, every time. A mixed approach produces a quantity nobody can reconcile.
Wall height assumed from floor-to-floor. Blockwork stops at the soffit; the beam depth is not blockwork. Measure the actual clear height per wall.
Finishes measured from the schedule instead of the plan. The schedule tells you *what*; only the plan tells you *how much*.

3 Structural Drawings

Structural drawings answer *what carries the load and how*. They are the source for concrete, reinforcement, formwork and structural steel — and the discipline where a misread symbol costs the most, because reinforcement is priced by weight and weight comes from bar diameter and length.



Typical beam section — read the bar mark, then the schedule, then the bending detail. Never scale from the drawing.

Beam section — the four things to read: overall size, bar mark and diameter, link diameter and spacing, and cover. The bar mark points to the bending schedule, which is the only place the cutting length exists.

SYMBOL / NOTATION	MEANING	TAKE-OFF CONSEQUENCE
`4T20`	4 bars, high-yield deformed, 20 mm diameter	Weight = 4 × length × 2.47 kg/m
`T10 @ 150 c/c`	10 mm links at 150 mm centres	Count = run length / 150 + 1, each cut to the link perimeter
`Ø12-200`	12 mm at 200 mm spacing (alternative notation)	Same calculation, different prefix — check the legend on each office's drawings
`Ld`	Development / anchorage length	Affects bar length beyond the clear span; do not price laps as zero
`C30/37`	Characteristic cylinder / cube strength	Governs the mix and therefore the rate, not only the quantity

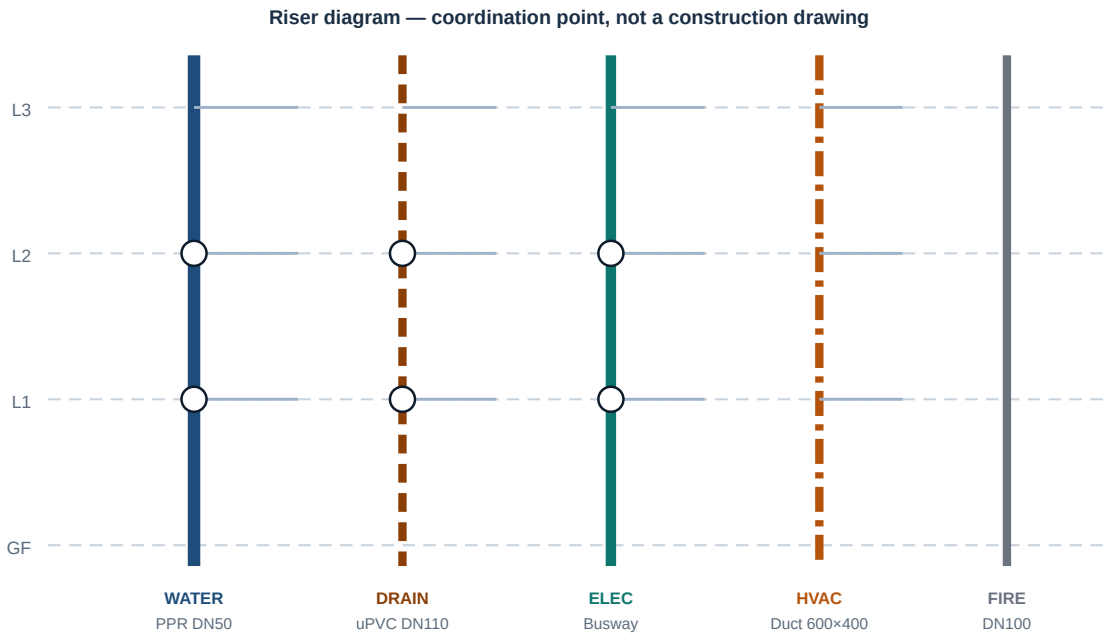
SYMBOL / NOTATION	MEANING	TAKE-OFF CONSEQUENCE
SL / EL	Structural level / finished floor level	Slab thickness = SL − soffit; screed sits above SL
BS / BW	Bottom steel / both ways	Bottom-only means top steel exists only over supports

Reading a reinforcement drawing in the right order

1. **Element, then mark:** identify the element (footing F1, column C2, beam B12) and its mark. Everything else references the mark.
2. **Section or schedule:** read the bar arrangement from the section, and the bar mark, diameter, shape and bending dimensions from the schedule.
3. **Cutting length:** from the bending schedule — including hooks, bends and laps. This is the only correct input for weight; a scaled length from the drawing omits every lap.
4. **Weight:** $$\text{kg/m} = \frac{d^2}{162}$$ for deformed bar — 12 mm gives 0.888 kg/m, 16 mm gives 1.578 kg/m, 20 mm gives 2.469 kg/m, 25 mm gives 3.854 kg/m.
5. **Links:** count along the member at the specified spacing, add the end zones where spacing is tightened, and cut to the perimeter less cover plus hooks.
6. **Laps and anchorage:** add them. A take-off that ignores laps under-measures reinforcement by roughly 3–8% depending on bar size and detailing.

4 MEP Drawings

MEP drawings answer *how the building services work and where they run*. Read them as systems, not as lines: a pipe on a plan means nothing until you know what it carries, what pressure it works at, and where it rises.



Riser diagram — the coordination document. Every horizontal service originates from a riser, so a riser that is too small is a project-wide problem, not a local one.

DISCIPLINE	DRAWING TYPES	KEY SYMBOLS	TAKE-OFF UNIT
Electrical	Single-line diagram, power layout, lighting layout, containment layout, earthing	DB, MCC, luminaire type, socket type, cable size and core count	Points, metres of cable and conduit, boards
Plumbing / drainage	Water riser, drainage riser, floor plans, details	Gate valve, check valve, floor drain, vent stack, clean-out	Metres by diameter, fittings by count, sanitary sets
HVAC	Duct layout, pipe sizing, equipment schedule, control diagram	AHU, FCU, VAV box, diffuser, damper, refrigerant pair	Metres of duct by size, metres of pipe, equipment count
Fire protection	Sprinkler layout, standpipe, pump room, alarm layout	Sprinkler head, alarm valve, flow switch, detector, sounder	Heads by count, pipe by diameter, devices by count
BMS / controls	Control schematics, points schedule, architecture diagram	Sensor, actuator, controller, points type (AI/AO/DI/DO)	Points by type — not metres

The services take-off rule

Measure pipe and duct by **centre-line length**, then add fittings separately. Never measure the drawn length of a pipe that turns: the drawing shows the run, the fittings schedule gives the elbows. A pipe take-off without fittings is typically 30–40% short of the installed cost.

5 Worked Example — A Simple Residential Building

A three-storey residential building, 12.0 × 10.0 m on a 6.0 m grid, 16 isolated footings, 32 columns, three suspended slabs. The full arithmetic is in Form TP-12; this is the reading method that produced it.

STEP	READ	COMPUTE	RESULT
1	Grid spacing 6,000 mm both ways; 4 bays × 3 bays	12.0 × 10.0 m footprint	120 m² per floor
2	Footing schedule F1: 1,500 × 1,500 × 400 deep, 16 no.	1.5 × 1.5 × 0.4 × 16	14.40 m³
3	Column schedule: 300 × 400, 3.2 m high, 32 no.	0.3 × 0.4 × 3.2 × 32	12.29 m³
4	Beam schedule: 250 × 500, total run 88 m per floor, 2 floors	0.25 × 0.5 × 88 × 2	22.00 m³
5	Slab: 120 m² × 150 mm, 3 floors	12 × 10 × 0.15 × 3	54.00 m³
6	Reinforcement: from the bar bending schedules, summed by diameter	Enter the schedule total, do not apply a percentage	11.248 t
7	Formwork: sides and soffits from the sections	Per element perimeter × length + soffit area	893.76 m²
8	Blockwork: external and internal wall lengths × clear height, less openings	Net area after deduction	522.48 m²
9	Plaster: both faces of every wall plus soffits	Net wall area × 2 + ceiling area	1,668.96 m²
10	Electrical: points from the power and lighting layouts, cable from the containment routes	Points by count; cable by route length × 1.15 for slack	14 no. / 1,850 m

> **Why the reinforcement figure is 11.248 t and not a percentage:** a percentage-of-concrete-volume estimate would give somewhere between 100 and 140 kg/m³, i.e. 10.3–14.5 t on 102.7 m³ — a range of ±20%. The schedule total is exact and defensible. Use the percentage only as a sanity check on the schedule, never as the quantity.

6 Cross-Checking the Set

CHECK	COMPARE	A DISCREPANCY MEANS
Slab thickness	Architectural section vs structural section	One discipline has not seen the other's revision
Ceiling height	Architectural RCP vs MEP ceiling void requirement	Services will not fit; ceiling level must drop
Opening sizes	Architectural door schedule vs structural opening	Frame will not fit, or excessive making-good
Levels	Architectural FFL vs structural SL	Screed thickness assumed incorrectly throughout
Column positions	Grid on plan vs grid on foundation layout	Setting-out error before excavation
Services routes	MEP plan vs structural beam depth	Beam cannot be penetrated where the drawing shows
Quantities	Drawing take-off vs BOQ	Either a measurement error or a BOQ omission — both are priced risks

7 Documenting Your Reading

Every take-off must be traceable back to a sheet, a revision and a dimension. Record, for each measured item: drawing number, revision, the element mark, and the dimensions used. When the drawing is revised, you can then identify exactly which quantities change — and claim the difference if it is a variation.

ITEM	DRAWING NO.	REV.	ELEMENT MARK	DIMENSION SOURCE	MEASURED BY	DATE

8 Common Errors and Their Cost

ERROR	HOW IT HAPPENS	TYPICAL COST CONSEQUENCE
Measuring a superseded revision	Using a file that was renamed, not reissued	Whole-element re-measure; often discovered after award
Scaling from a PDF	No printed dimension available for a particular element	1–3% random error per dimension, compounding across elements
Ignoring laps and anchorage	Using clear bar length from the drawing	3–8% under-measure of reinforcement
No deduction for openings	Measuring wall area as a solid rectangle	2–6% over-measure of blockwork and plaster
Forgetting the waste factor	Pricing net quantity as the purchase quantity	3–10% under-cost depending on the material
Counting fittings as pipe	Taking a centre-line length and stopping there	30–40% under-cost on plumbing packages
Assuming floor-to-floor wall height	Blockwork measured to slab soffit including beam depth	Over-measure of 8–15% on blockwork

ERROR	HOW IT HAPPENS	TYPICAL COST CONSEQUENCE
Missing a discipline entirely	Fire alarm or BMS absent from the drawing register	Whole trade missing from the price

Prepared by

Technical manager

Reviewed by

Chief estimator

FORM TP-14 · QUANTITY TAKE-OFF & SUMMARY SHEET

Quantity Take-off Sheet and Summary Sheet

Dimensions in, quantities out. Every figure below is computed from the length, width, height, number and waste percentage in its own row — change a dimension and the quantity, the group subtotal and the total all follow.

ITEMS	GROUPS
24	9

SUMMARY	REV.
Linked	A

> **This sheet is the source; the BOQ is the destination.** Quantities measured here must appear unchanged in Form TP-09. If the two disagree, one of them is wrong and the evaluator will find it. The cross-check at the foot of this page must agree with TP-09 to the cent.

1 Sheet Particulars

PROJECT

DRAWING SET / REVISION

MEASURED BY

DATE

ELEMENT / PACKAGE

UNITS

m, m², m³, t, no.

WASTE POLICY APPLIED

By material

CHECKED BY

2 Take-off Sheet

Net = L × W × H × No. · Gross = Net × (1 + waste%). · Waste is applied per row because it differs by material: rebar 4%, concrete 3%, blockwork 6%, tiles 10%. · Value = Gross × the BOQ rate carried from TP-09.

CODE	DESCRIPTION OF WORKS	UNIT	L	W	H / D	NO.	NET QTY	WASTE %	GROSS QTY	VALUE
01 · EARTHWORKS & BACKFILL										
01-01	Excavation for foundations	m³	2.2	2.2	1.5	16	116.160	0	116.160	$ 1,068.67
01-02	Backfilling & compaction	m³	81.2	1	1	1	81.296	10	89.426	$ 380.06
02 · CONCRETE WORKS										
02-01	Blinding concrete C15	m³	1.7	1.7	0.1	16	4.624	5	4.855	$ 582.62
02-02a	Isolated footings C30	m³	1.5	1.5	0.4	16	14.400	3	14.832	$ 2,343.46
02-02b	Tie beams C30	m³	0.3	0.6	88	1	15.840	3	16.315	$ 2,577.80
02-02c	Columns C30	m³	0.3	0.4	3.2	32	12.288	3	12.657	$ 1,999.75
02-02d	Beams C30	m³	0.25	0.5	88	2	22.000	3	22.660	$ 3,580.28
02-02e	Slabs C30	m³	12.0	10.0	0.15	3	54.000	3	55.620	$ 8,787.96
03 · REINFORCEMENT										
03-01	High-yield reinforcement B420	t	11.2	1	1	1	11.248	4	11.698	$ 11,405.72
04 · FORMWORK										
04-01	Formwork — sides, soffits and slabs	m²	893.	1	1	1	893.760	5	938.448	$ 5,959.14
05 · MASONRY										
05-01	Lightweight blockwork	m²	522.	1	1	1	522.480	6	553.829	$ 11,353.49
06 · PLASTER & PAINT										
06-01	Cement plaster	m²	166	1	1	1	1,668.960	8	1,802.477	$ 13,969.20
06-02	Emulsion paint	m²	166	1	1	1	1,668.960	5	1,752.408	$ 7,885.84
07 · TILING										
07-01	Floor tiles 600×600	m²	12.0	10.0	1	2	240.000	10	264.000	$ 6,705.60
07-02	Wall tiles	m²	15.0	1	1	12	180.000	10	198.000	$ 4,474.80
07-03	Skirting	m	150	1	1	1	150.000	7	160.500	$ 2,038.35
08 · ELECTRICAL										
08-01	Lighting & socket points	no.	14	1	1	12	168.000	2	171.360	$ 5,312.16
08-02	Cable installations	m	185	1	1	1	1,850.000	8	1,998.000	$ 10,689.30
08-03	Conduits & trunking	m	160	1	1	1	1,600.000	8	1,728.000	$ 3,888.00
08-04	Distribution boards	no.	4	1	1	1	4.000	0	4.000	$ 820.00
Total measured value at BOQ rates										$ 117,014.31

CODE	DESCRIPTION OF WORKS	UNIT	L	W	H / D	NO.	NET QTY	WASTE %	GROSS QTY	VALUE
09 · MECHANICAL & PLUMBING										
09-01	Copper A/C refrigerant pipework	m	220	1	1	1	220.000	6	233.200	$ 4,104.32
09-02	PVC drainage	m	120	1	1	1	120.000	7	128.400	$ 1,521.54
09-03	PPR water supply	m	260	1	1	1	260.000	7	278.200	$ 2,434.25
09-04	Sanitary sets	set	12	1	1	1	12.000	0	12.000	$ 3,132.00
Total measured value at BOQ rates										$ 117,014.31

3 Summary Sheet

Group subtotals carried to the BOQ. The share column is what tells you where to focus a pricing review: a group above 20% of the total deserves a second check of both quantity and rate.

WORK GROUP	VALUE AT BOQ RATES	SHARE
Earthworks & backfill	$ 1,448.73	1.2%
Concrete works	$ 19,871.87	17.0%
Reinforcement	$ 11,405.72	9.7%
Formwork	$ 5,959.14	5.1%
Masonry & blockwork	$ 11,353.49	9.7%
Plaster & paint	$ 21,855.04	18.7%
Tiling & flooring	$ 13,218.75	11.3%
Electrical	$ 20,709.46	17.7%
Mechanical & plumbing	$ 11,192.11	9.6%
Total	**$ 117,014.31**	**100%**

CROSS-CHECK	TP-14	TP-09
Total direct cost	$ 117,014.31	117,014.31
After overhead 12%	$ 131,056.03	131,056.03
Offer value	$ 141,540.51	141,540.51

If the two columns disagree

The difference is almost always one of: a row measured here but not priced in the BOQ (an omission), a row priced in the BOQ but not measured here (a duplicate), or a different waste percentage in the two places. Find the row before you change a total.

4 Waste Analysis

MATERIAL	WASTE APPLIED	BASIS	NOTE
Reinforcement	4%	Cutting and lap allowance	Add laps separately; this covers offcut only
Concrete	3%	Over-pour and spillage	Higher on small sections and pumped pours
Formwork	5%	Damage and re-use loss	Depends on the number of re-uses planned
Blockwork	6%	Cutting at openings and corners	Rises with the number of openings
Plaster	8%	Thickness variation and waste	Applies to the mortar, not the area
Paint	5%	Application loss	Per coat — multiply by the number of coats
Tiles	10%	Cutting and pattern	Diagonal or patterned layouts need 15%
Pipe and cable	6–8%	Slack, bends and terminations	Cable also needs a route-length factor, not only waste

5 Verification of the Worked Example

The figures on this sheet come from the worked example in Guide TP-G1: a 12.0 × 10.0 m residential building on a 6.0 m grid, 16 footings, 32 columns, three suspended slabs. Two independent checks:

Check 1 — concrete by element

Blinding 1.70×1.70×0.10×16	4.624 m³
Footings 1.50×1.50×0.40×16	14.400 m³
Tie beams 0.30×0.60×88	15.840 m³
Columns 0.30×0.40×3.20×32	12.288 m³
Beams 0.25×0.50×88×2	22.000 m³
Slabs 12×10×0.15×3	54.000 m³
Sum	**123.15 m³**

Check 2 — reinforcement per cubic metre

11.248 t over 123.15 m³ of structural concrete (excluding blinding, which is unreinforced) gives **91 kg/m³**. For a residential frame this sits inside the normal 90–130 kg/m³ band — low but not implausible for light slabs on a regular grid.

A result outside 80–150 kg/m³ on a building of this type means the schedule has been misread, a diameter omitted, or laps left out. Use this ratio as a red flag, never as the quantity.

Measured by	Checked by	Approved by
Quantity surveyor	Chief estimator	Commercial manager

GUIDE TP-G2 · HOW TO USE THIS KIT

How to Use the Kit: Filling the Forms, Pricing, and Reviewing Before Submission

The order of work matters as much as the content. Fill in this sequence and the documents will agree with each other.

SECTIONS	FORMS
7	13

SHEETS	REV.
7	A

Read this page first. The kit is not thirteen independent forms; it is one offer expressed thirteen ways. The quantities flow from TP-14 into TP-09, the payment percentages in TP-03 follow from the BOQ groups, and the methodology in TP-02 must match both. Fill them out of order and they will contradict each other in front of the evaluator.

1 What Is in the Kit

CODE	DOCUMENT	FILL IT WHEN	FEEDS
TP-01	Cover letter & compliance statement	Last, before sealing	The whole offer
TP-02	Scope understanding & methodology	First	TP-03, TP-04, TP-05
TP-03	Time schedule & payment schedule	After TP-09 groups are known	Financial offer
TP-04	QA/QC plan	Alongside TP-02	Closeout obligations
TP-05	HSE plan	Alongside TP-02	Site mobilisation
TP-06	Resource plan (org, CVs, plant, materials)	Early — it takes the longest	TP-03 credibility
TP-07	Track record	Early — certificates must be requested	Prequalification score
TP-08	Risk register	After TP-02	Pricing contingency
TP-09	BOQ summary & price schedule	After TP-14	Financial offer, TP-03
TP-10–13	Tender checklist, RFI, deviations, handover warranty	Checklist first, warranty at handover	Submission completeness
TP-14	Quantity take-off & summary	Before TP-09	TP-09
TP-G1	Drawing reading guide	While measuring	TP-14
TP-G2	This guide	Before starting	—

2 The Order of Work

1 **Read the tender documents completely, once, before writing anything**

Instructions, conditions, specification, drawings, BOQ and every addendum. Then read the drawing register and confirm which revisions are current. An offer built on a superseded drawing is worthless however well written.

2 **Complete TP-10, the tender checklist**

This tells you what must be produced and what must be requested from other people — certificates, references, audited accounts. Those take days; start them now.

3 **Measure, using TP-G1 and TP-14**

Every quantity traced to a drawing number, revision and dimension. Do not price anything until the take-off is complete, or you will price the same item twice.

4 **Price in TP-09**

Enter unit rates, apply waste and the overhead and profit factors. The offer value emerges from the arithmetic — do not decide the offer value first and back-fill rates.

5 **Write TP-02, then TP-04 and TP-05**

The methodology must describe the project you have just measured. Specificity here is what separates a scored submission from a rejected one.

6 **Fill TP-06, TP-07 and TP-08**

Names, plant, sources, references and risks. Each must be verifiable, and none may be repeated from another tender without checking it is still true.

7 **Build TP-03 from the priced BOQ**

Payment percentages must follow the value of the work, not a preferred cash flow. If the programme and the payment ladder disagree, the evaluator sees it immediately.

8 **Complete TP-12 (deviations) and TP-11 (any outstanding RFIs)**

Every departure recorded before submission; anything unresolved becomes your risk at the price you bid.

9 **Write TP-01 last and sign everything**

The cover letter summarises an offer that now exists. Check that the figure in the letter matches TP-09 to the last digit.

3 Linking Quantities to Prices

The chain is deliberately linear, and each link is checkable:

STEP 1

Measure

TP-14 · L × W × H × No., then waste

STEP 2

Summarise

TP-14 summary · 9 group subtotals

STEP 3

Price

TP-09 · rate per item, waste per row

STEP 4

Build up

Overhead 12% → profit 8% → tax

STEP 5

Distribute

TP-03 · payment % per milestone

The single most common internal inconsistency

A payment schedule written before the BOQ was priced. It always over-weights early milestones, because mobilisation feels expensive when you are staring at an empty site. It is not: on a building project, mobilisation is typically 3–5% of the contract value. If TP-03 claims 15% at mobilisation, the evaluator will query it, and if it is accepted the employer will query it again at the first payment certificate.

4 Filling the Forms in a Browser

- **Type directly into the page.** Every field is editable and saves to your own browser as you type — close the tab and come back, the values are still there. They are not sent anywhere.
- **Print or save as PDF** with the button in the top bar. Use your browser's print dialog, A4, and enable background graphics so the shaded tables print correctly.
- **Clear** wipes the saved values for the current document only, after a confirmation prompt. Use it when you start a new tender.
- **Currency** switches every money figure between USD, JOD, SAR and AED at fixed reference rates. Edit the rates in `assets/en.js` if you need live rates — the figures are converted at display time, so nothing you typed is destroyed.
- **Never print to a scanned image** and then fill by hand; it defeats the calculation and loses the record trail.

5 Pre-Submission Review

Three reviews, by three different people, in this order. A single reviewer catches layout errors; three catch arithmetic, omissions and inconsistencies.

REVIEW	REVIEWER	CHECKS	SIGNATURE
1 — Technical	Technical manager / chief engineer	Scope complete, methodology matches the drawings, no discipline missing, TP-02 consistent with TP-04 and TP-05	
2 — Commercial	Quantity surveyor / commercial manager	Every BOQ item priced, no zero rates, waste applied once, TP-14 and TP-09 agree to the cent, offer value matches the financial letter	
3 — Administrative	Contracts / proposals manager	TP-10 checklist complete, all signatures and stamps, bonds and certificates current, additions acknowledged, copies counted, deadline confirmed	

FINAL CONSISTENCY CHECKS — TICK EVERY ONE

#	CHECK	WHERE	✓
1	BOQ total equals the TP-14 summary total	TP-09 / TP-14	
2	Offer value in TP-01 equals the financial offer letter	TP-01	
3	Payment percentages in TP-03 sum to 100%	TP-03	
4	Milestone values are defensible from the BOQ group shares	TP-03 / TP-09	
5	No item has a zero or blank rate	TP-09	
6	Waste applied once, not twice	TP-09 / TP-14	
7	Every deviation appears in TP-12 as well as TP-01	TP-12	
8	Programme duration matches the stated construction period	TP-03	
9	Key personnel not double-booked against another live tender	TP-06	
10	Every certificate cited in TP-07 is attached and current	TP-07	
11	All addenda acknowledged	TP-10	
12	Bid bond amount, form and validity correct	TP-10	
13	Signed, stamped, dated on every form that requires it	All	
14	Currency stated consistently throughout	All	
15	Correct number of copies, original marked	Submission	

6 Common Reasons Offers Are Rejected

REASON	HOW IT HAPPENS	HOW THIS KIT PREVENTS IT
Missing mandatory document	Checklist not used; certificate expired	TP-10 with an owner and status per line
Unsigned or unstamped form	Signed the PDF, not the page	Signature blocks on every form that needs one
Arithmetic inconsistency between documents	Offer value typed by hand into three places	One calculated value; TP-14 and TP-09 cross-check to the cent
Unqualified departure	Deviation stated in the cover letter only	TP-12 deviations schedule, cross-referenced from TP-01

REASON	HOW IT HAPPENS	HOW THIS KIT PREVENTS IT
Programme inconsistent with the price	Payment ladder written before pricing	TP-03 built from the priced BOQ groups
Generic methodology	Reused from another tender	TP-02 requires project-specific content per package
Late submission	Courier relied on to meet the deadline	Checklist deadline line, administrative review sign-off

7 What to Change Before You Submit

Mandatory substitutions

Every unit rate in TP-09 is **illustrative**. It exists to demonstrate the arithmetic of the chain from take-off to offer value, and it is not a market price. Before submission, replace every rate with a live supplier or subcontract quotation, and replace the overhead and profit factors with your own company figures. The quantities may be used as a worked example; the prices may not.

Checklist for adapting the example

- Replace the project name, client, tender number and dates on TP-01.
- Replace the programme durations on TP-03 with your own planned durations.
- Replace every CV, project name, client and reference on TP-06 and TP-07 with real, verifiable ones.
- Replace the risk register entries on TP-08 with risks specific to the actual site and scope.
- Replace all 24 rates on TP-09 and the overhead and profit factors.
- Adjust TP-14 dimensions to the actual drawings, then re-check that TP-14 and TP-09 still agree.

Prepared by

Proposals manager

Approved for use

General manager